PILZ SAFETY ARCHITECTURE - 11 LOCKED GUARD DOORS

PLC / HMI

S7-1512C-1 PN

PROFINET

PNOZmulti 2 SAFETY SYSTEM

PNOZ m ES PROFINET	PNOZ m B0	PNOZ m EF 16DI	PNOZ m EF 8DI4DO	PNOZ m EF 1MM
772138	772100	772140	772142	772170

SAFE STOP

SAFE STOP OUTPUT

O0 + O1
S1 + S2 force-guided
EDM monitored
STO / hazardous power OFF

SAFE UNLOCK PERMISSION

2 safe outputs
KUL1 + KUL2 force-guided
EDM monitored
Creates SAFE_UL_24V

SAFE UNLOCK ENABLE

PILZ PRODUCT SPECIFICATION

772138 PNOZ m ES PROFINET - PROFINET IO, 2 x RJ45 (non-safe communication)
772100 PNOZ m B0 - base unit, 20 safe/configurable inputs, 4 safe semiconductor outputs
772140 PNOZ m EF 16DI - 16 safe digital inputs
772142 PNOZ m EF 8DI4DO - 8 safe digital inputs + 4 safe semiconductor outputs
772170 PNOZ m EF 1MM - safe speed/standstill monitoring, 1 axis

DUAL-CHANNEL SAFE INPUTS: 21-22 (GUARD) + 31-32 (LOCK)

GS1	GS2	GS3	GS4	GS5	GS6	GS7	GS8	GS9	GS10	GS11
FRONT 1	FRONT 2	FRONT 3	FRONT 4	REAR 1	REAR 2	REAR 3	REAR 4	RIGHT 1	RIGHT 2	LEFT 1

GS1-GS8 -> PNOZ m EF 16DI (I20-I35)

GS9-GS11 -> PNOZ m EF 8DI4DO (I40-I45); I46-I47 SPARE

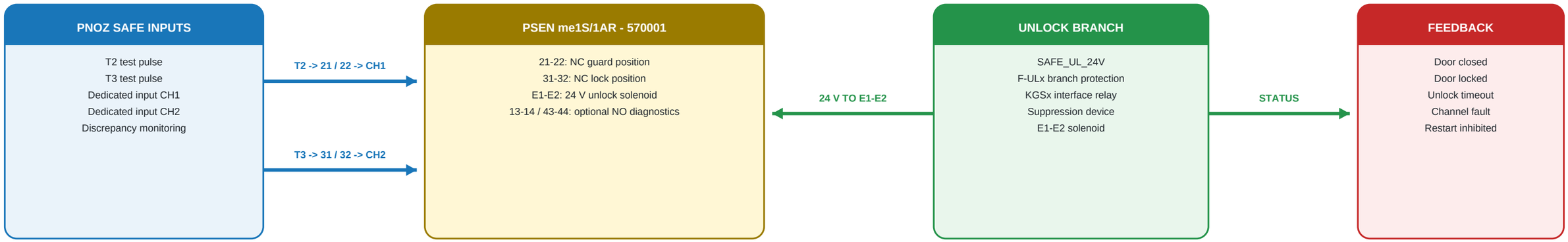
SAFE_UL_24V - INDIVIDUAL BRANCHES F-UL1...11 / KGS1...11 / E1-E2 COILS

F-UL1 + KGS1	F-UL2 + KGS2	F-UL3 + KGS3	F-UL4 + KGS4	F-UL5 + KGS5	F-UL6 + KGS6	F-UL7 + KGS7	F-UL8 + KGS8	F-UL9 + KGS9	F-UL10 + KGS10	F-UL11 + KGS11

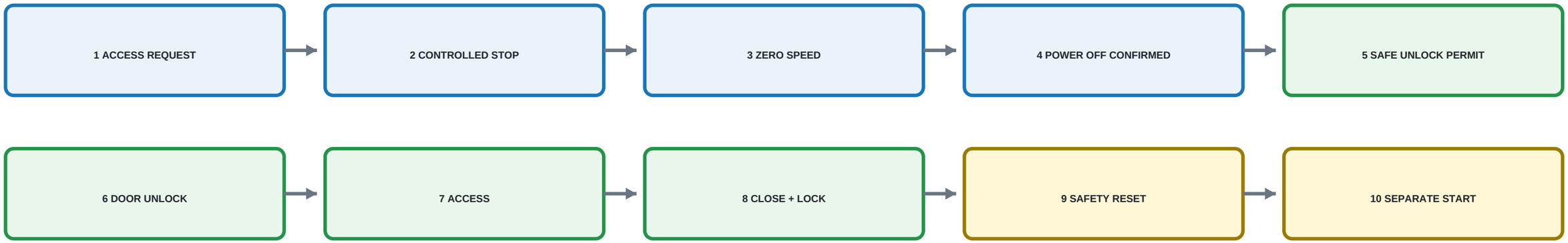
PROFINET / safe unlock power
Dual-channel safety input
Safe stop / STO
Individual relay / guard lock

NOT FOR CONSTRUCTION - Safety validation, stop time, PLr and unlock zoning must be approved before wiring.

TYPICAL GUARD CONNECTION - REPEAT FOR GS1 TO GS11



Safe operating sequence



MANDATORY: no automatic restart. Any open/unlocked guard, channel discrepancy, EDM fault or loss of safety communication inhibits restart.

Electrical drawing must show XGS1-XGS11 terminal blocks, wire/core numbers, E1-E2 connections, individual protection and cross-references to the PNOZmulti project.

All eleven device nameplates and the second PNOZ expansion-module nameplate must be verified before issue for construction.